

COURSE STRUCTURE

Course Code	MCA40190				
Course Category	PR				
Course Title	Lab on Advanced Database Management System				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours			2		2
Credits			1		1
Assessment Schema Code	PJ				
<u>Prerequisites:</u>					
Database Management System and Relational database management system, SQL					
<u>Course Objectives:</u>					
1. Understanding parallel database principles, distributed, XML, Objective oriented and mobile databases.					
<u>Course Outcomes:</u>					
After completion of this course students will be able to: 1. Design parallel and distributed databases. 2. To work on mobile, XML databases.					
<u>Course Contents:</u>					
Sample Lab Assignments					
1. Write down Stored Procedure for inserting, updating values in the table					
2. Write down stored procedures to accept input values as a parameter and update values of the tables					
3. Write down DML trigger to raise the error while inserting duplicate value in the table					
4. Write down DML trigger to raise the error violating Check Constraints					
5. Write down DML trigger to raise the error when user deletes more than records from table					
6. Create index and measure the performance of query on the table					
7. Compare results before and after indexing by considering any sample table					

8. Write down the stored procedure for selecting the first five records and copy those five records in another table
9. Write down a function which prints higher salaried person from table by inputting his empid and name
10. Write down the function which selects part name (substring, like surname, firstname or middle name and copy that part in another table's column)

Laboratory Exercises / Practical:

Every student will carry out minimum *Six Practical*/exercises and submit the journal, which will be evaluated as part of continuous assessment.

Learning Resources:

Reference Books:

1. AviSilberschatz, Henry F. Korth, S. Sudarshan, "Database System Concept", 4th Edition, 2001, ISBN: 0072283637, McGraw-Hill publications.
2. RamezElmasri, ShamkantNavathe, "Fundamental of Database Systems", 5th Edition, 2009, ISBN: 9780321369574, Pearson Education.
3. Alexis Leon, Mathews Leon, "Database Systems concept", 2002, ISBN: 9788125911654, Leon Vikas.

Supplementary Reading:

Web Resources:

1. <https://www.tutorialspoint.com> ,
2. www.w3schools.com

MOOCs: Online courses for self-learning

Courses by NPTEL and MIT Open Courseware etc.

https://onlinecourses.nptel.ac.in/noc22_cs47/preview

Pedagogy:

- Power point presentations
- Videos
- Demonstrations
- Interactive discussion.
- Practical based learning,
- Project based learning,